

1. a) Zapiši eksplicitno $(\operatorname{Re}(), \operatorname{Im}())$

$$\cos(x+iy) = \frac{1}{2}(e^{i(x+iy)} + e^{-i(x+iy)}) = \frac{1}{2}(e^{-y}\cos x + ie^{-y}\sin x + e^y\cos x - ie^y\sin x) =$$

$$= \dots = \underline{\cos x \cosh y - i \sin x \sinh y}$$

$$\sin(x+iy) = \frac{1}{2i}(e^{i(x+iy)} - e^{-i(x+iy)}) = -i(\cos x \sinh y + i \sin x \cosh y) = \underline{\sin x \cosh y + i \cos x \sinh y}$$

b) Preveri, da sta $\sin, \cos$ holomorfni

CR enačbi: $u_x = v_y, u_y = -v_x$

$$\cos(x+iy) = \underbrace{\cos x \cosh y}_u - i \underbrace{\sin x \sinh y}_v$$

$$u_x = \sin x \cosh y \quad \checkmark$$

$$v_y = -\sin x \cosh y \quad \checkmark$$

$$u_y = \cos x \sinh y \quad \checkmark$$

$$-v_x = \cos x \sinh y \quad \checkmark$$

$\sin(x+iy)$ podobno

c) Izračunaj odvoda

$$\cos'(x+iy) = u_x + i v_x = -\sin x \cosh y - i \cos x \sinh y = \underline{-\sin(x+iy)}$$

$$\sin'(x+iy) = \underline{\cos(x+iy)}$$

2. $f: D \rightarrow \mathbb{C}$ nekajst. holomorfna funkc.

Ali je $f_1(z) = f(\bar{z})$ tudi holomorfna?

$$f(x+iy) = u(x,y) + i v(x,y) \quad \rightarrow \quad f_1(x+iy) = f(x-iy) = u(x,-y) + i v(x,-y)$$

$$\underbrace{u_x = v_y, u_y = -v_x}_{\text{a)}}$$

$$U(x,y) + V(x,y)$$

$$U_x(x,y) = u_x(x,-y); \quad U_y(x,y) = -u_y(x,-y)$$

$$V_x(x,y) = v_x(x,-y); \quad V_y(x,y) = -v_y(x,-y)$$

Ali velja $U_x = V_y$ in $U_y = -V_x$?

$$u_x(x, -y) = -v_y(x, -y) \quad u_y(x, -y) = v_x(x, -y)$$

②

$$u_x(x, -y) = -u_x(x, -y) \quad u_y(x, -y) = -u_y(x, -y)$$

$\Rightarrow u_x = 0, u_y = 0 \Rightarrow f_1$ je holomorfná, če u in v konstantni

③. $f: D \rightarrow \mathbb{C}$ holomorfná; $f_2(z) = \overline{f(\bar{z})}$ Ali je f_2 holomorfná?

$$f_2(x+iy) = U(x, y) + iV(x, y) = \overline{f(x-iy)} = u(x, -y) - i v(x, -y)$$

$$U_x = u_x \quad V_x = -v_x \quad u_x = v_y \Rightarrow U_x = V_y \checkmark$$

$$U_y = -u_y \quad V_y = v_y \quad u_y = -v_x \Rightarrow U_y = -V_x \checkmark$$

④. $f: D \rightarrow \mathbb{C}$ holomorfná, $|f(z)| = \text{konst.}$ Pokaži, da je f konstantna

Vrste

$f: B(z_0, r) \rightarrow \mathbb{C}$ holomorfná $\Rightarrow f(z) = \sum_{n=0}^{\infty} c_n (z-z_0)^n$; ima konv. radij vsaj r
odprt krog

$$c_n = \frac{f^{(n)}(z)}{n!} = \frac{1}{2\pi i} \oint \frac{f(z)}{(z-z_0)^{n+1}} dz$$

$\gamma \leftarrow r$ u obkroži z_0 v poz. smeri

Laurentove vrste

$$K = \{z \in \mathbb{C}; r < |z - z_0| < R\} \text{ kolobar}$$

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n; \quad c_n = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z - z_0)^{n+1}} dz$$

$$f(z) = \underbrace{\dots + \frac{c_{-2}}{(z - z_0)^2} + \frac{c_{-1}}{(z - z_0)} + c_0}_{\text{glavni del}} + c_1(z - z_0) + \dots$$

① Razvij $f(z) = z^2 e^{1/z}$ v Laurentovo vrsto na kolobarju $\mathbb{C} - \{0\}$.

$$e^z = 1 + z + \frac{z^2}{2!} + \dots \quad \forall z \in \mathbb{C}$$

$$e^{1/z} = 1 + z^{-1} + \frac{z^{-2}}{2!} + \dots \quad \forall z \in \mathbb{C} \setminus \{0\}$$

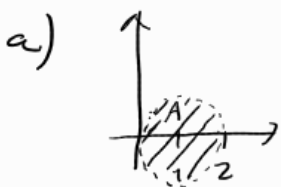
$$\rightarrow f(z) = z^2 + z + \frac{1}{2!} + \underbrace{\frac{z^{-1}}{3!} + \dots}_{\text{glavni del vsebuje } \infty \text{ členov}}$$

$\Rightarrow z=0$ je bistvena singularnost

② razvij $f(z) = \frac{1}{(z-1)(z-2)}$ v Laurentovo vrsto, ki bo konv. na:

a) $A = \{z \in \mathbb{C}; 0 < |z-1| < 1\};$

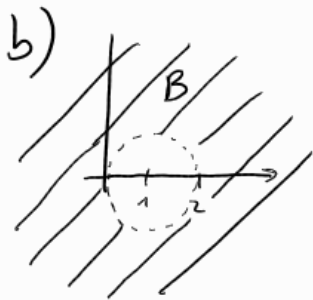
b) $B = \{z \in \mathbb{C}; |z-1| > 1\}.$



$$f(z) = \frac{1}{z-1} \left(\frac{1}{z-2} \right) = \frac{1}{z-1} \cdot \frac{1}{(z-1)-1} = -\frac{1}{z-1} \left(\frac{1}{1-(z-1)} \right) =$$
$$= -\frac{1}{z-1} \left(1 + (z-1) + (z-1)^2 + \dots \right) =$$

$$= -\frac{1}{z-1} - 1 - (z-1) - (z-1)^2 - \dots$$

$\downarrow$
 $z_0=1$ je pol 1. st.; konv. na A



$$f(z) = \frac{1}{z-1} \stackrel{!}{=} \frac{1}{(z-1)-1} = \frac{1}{z-1} \frac{1}{1 - \frac{1}{z-1}} = \frac{1}{z-1} < 1; \text{ razvijemo v geom. vrsto}$$

$$= \frac{1}{(z-1)^2} \left(1 + \frac{1}{z-1} + \frac{1}{(z-1)^2} + \dots \right) =$$

$$= \dots + \frac{1}{(z-1)^3} + \frac{1}{(z-1)^2}$$

Liouville izrek

$f: \mathbb{C} \rightarrow \mathbb{C}$ holomorfna, $|f(z)| \leq M \Rightarrow f$ konstantna

$f: \mathbb{C} \rightarrow \mathbb{C}$ holomorfna, $\operatorname{Im}(f(z)) > 0 \quad \forall z \in \mathbb{C}$. Pokaži, da je f holomorfna

$$f(z) = u + iv, \quad v > 0$$

$$u_x = v_y$$

$$u_y = -v_x$$

$$g(z) = e^{if(z)} = e^{iu-v} = e^{iu} e^{-v} \rightarrow |g(z)| = e^{-v} < 1 \Rightarrow g \text{ konstantna}$$

ker je e^z lokalno nekonstantna $\Rightarrow f(z)$ je konstantna ✓

Princip identičnosti

$f, g: D \rightarrow \mathbb{C}$ holomorfni, $A \subseteq D$ množica s stekališčem.

Če $f|_A = g|_A \Rightarrow f = g$ na D .

Poišči $f: \mathbb{C} \rightarrow \mathbb{C}$ holomorfna, da je $f\left(\frac{1}{n}\right) = e^{\frac{n^2-3n+2}{n^3}}; n \in \mathbb{N}$.

$f(z) = e^z e^{-3z^2} e^{2z^3}$ holomorfna, na $A = \left\{\frac{1}{n}; n \in \mathbb{N}\right\}$ se ujema z $f\left(\frac{1}{n}\right) \Rightarrow \checkmark$

Residuum

f holomorfna na punktirani okolici z_0 ($0 < |z - z_0| < r$),

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n \quad (\text{Laurentova vrsta}) \quad \text{na ok. } z_0$$

$$\rightarrow \text{Res}(f, z_0) = c_{-1}$$

$$\rightarrow z_0 \text{ pol stopnje } n: \text{Res}(f, z_0) = \lim_{z \rightarrow z_0} \frac{1}{(n-1)!} \left((z - z_0)^n f(z) \right)^{(n-1)} \quad \text{odvod}$$

$$f(z) = \frac{e^z}{\sinh(z)}, \quad \text{Res}(f, 0) = ?$$

$$\lim_{z \rightarrow 0} z \cdot f(z) = \lim_{z \rightarrow 0} \frac{z e^z}{\sinh z} = \lim_{z \rightarrow 0} \frac{z}{\sinh z} \cdot \lim_{z \rightarrow 0} e^z \stackrel{\text{L'H.}}{=} \lim_{z \rightarrow 0} \frac{1}{\cosh z} = 1 \quad \checkmark$$

$f: D \setminus \{z_1, \dots, z_n\} \rightarrow \mathbb{C}$ holomorfna, ∂D poz. orientiran

$$\oint_{\partial D} f dz = 2\pi i \sum_{k=1}^n \text{Res}(f, z_k)$$

$$\oint_{|z|=3} \frac{e^z}{z(1-z)^3} dz = ?$$

$$z=1: n=3$$

$$z=0: n=1$$

$$\text{Res}(f, 0) = \lim_{z \rightarrow 0} \frac{e^z}{(1-z)^3} = 1$$

$$\begin{aligned} \text{Res}(f, 1) &= \frac{1}{2} \lim_{z \rightarrow 1} \left(\frac{e^z}{z} \right)'' = \frac{1}{2} \lim_{z \rightarrow 1} \left(\frac{e^z z - e^z}{z^2} \right)' = \frac{1}{2} \lim_{z \rightarrow 1} \left(\frac{e^z z - e^z}{z^2} - \frac{e^z z^2 - 2ze^z}{z^3} \right) \\ &= \frac{1}{2} (0 - (-e)) = -\frac{e}{2} \end{aligned}$$

$$\rightarrow \oint_{|z|=3} f(z) dz = \left(1 - \frac{e}{2}\right) 2\pi i$$

Računanje $\mathbb{R}$ integrala s pomočjo residuov

$$1) \int_0^{2\pi} f(\sin x, \cos x) dx = \oint_{|z|=1} f\left(\frac{z-\frac{1}{z}}{2i}, \frac{z+\frac{1}{z}}{2}\right) \frac{dz}{iz}$$

$$\cos x = \frac{z+\frac{1}{z}}{2},$$

$$\sin x = \frac{z-\frac{1}{z}}{2i}$$

$$dx = \frac{dz}{iz}$$

$$\int_0^{2\pi} \frac{dx}{5-4\cos x} = \oint_{|z|=1} \frac{dz}{iz(5-4\frac{z+\frac{1}{z}}{2})} = \oint_{|z|=1} \frac{dz}{iz(5-2z-\frac{2}{z})} = \oint_{|z|=1} \frac{dz}{i(5z-2z^2-2)} = \oint_{|z|=1} \frac{idz}{2z^2-5z+2} =$$

$$= \oint_{|z|=1} \frac{idz}{2(z-2)(z-\frac{1}{2})} = 2\pi i \operatorname{Res}\left(\frac{i}{2(z-2)(z-\frac{1}{2})}, \frac{1}{2}\right) =$$

$$z_{1,2} = \frac{5 \pm \sqrt{25-4}}{2} = \begin{matrix} 2 \\ \frac{1}{2} \end{matrix}$$

$$= -2\pi \frac{1}{z(\frac{1}{2}-2)} = \underline{\underline{\frac{2\pi}{3}}}$$

$$2) \int_{-\infty}^{\infty} f(x) dx = \lim_{R \rightarrow \infty} \int_{-R}^R f(x) dx$$

f analitična $\Rightarrow f(z)$ holomorfná

$\rightarrow$ odvisno od primera:



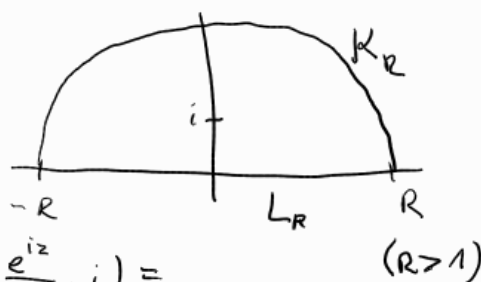
$$\oint_C f(z) dz = 2\pi i \sum \operatorname{Res}(f, \dots)$$

$$\int_{-R}^R f(x) dx + \int_{\Gamma_R} f(z) dz$$

$\uparrow$
 lok

$\underbrace{\int_{\Gamma_R} f(z) dz}_{\text{pogosta } R \rightarrow \infty \rightarrow 0}$

$$1) \text{ Izračunaj } \int_{-\infty}^{\infty} \underbrace{\frac{\cos x}{1+x^2}}_{f(x)} dx.$$

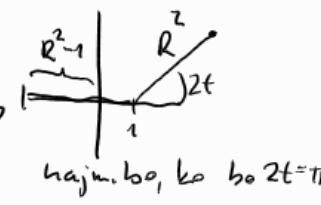


$$f(z) = \frac{e^{iz}}{1+z^2}$$

$\downarrow$
 bolje e^{iz} kot $\cos z$

$$\oint_{L_R \cup K_R} \frac{e^{iz}}{1+z^2} dz = 2\pi i \operatorname{Res}\left(\frac{e^{iz}}{1+z^2}, i\right) =$$

$$z=i \text{ pol } 1. \text{ stopnje} \rightarrow 2\pi i \lim_{z \rightarrow i} \frac{(z-i)e^{iz}}{(z+i)(z-i)} = 2\pi i \frac{e^{-1}}{2i} = \frac{\pi}{e}$$

$$\frac{\pi}{e} = \int_{-R}^R \frac{e^{ix}}{1+x^2} + \int_{K_R} \frac{e^{iz}}{1+z^2} dz = \int_{-R}^R \frac{\cos x}{1+x^2} + i \underbrace{\int_{-R}^R \frac{\sin x}{1+x^2}}_{\text{liha} \rightarrow 0} + \int_0^\pi \frac{e^{iRe^{it}} Re^{it}}{1+R^2 e^{2it}} dt$$


hajm. bo, ko bo $2t = \pi$

$$\left| \int_0^\pi \frac{e^{iRe^{it}} Re^{it}}{1+R^2 e^{2it}} dt \right| \leq \int_0^\pi \left| \frac{e^{iRe^{it}} Re^{it}}{1+R^2 e^{2it}} \right| dt = \int_0^\pi \frac{R |e^{i \cos t - \sin t}|}{|1+R^2 e^{2it}|} dt \leq \int_0^\pi \frac{R e^{-\sin t}}{R^2 - 1} dt \leq \frac{R}{R^2 - 1} \int_0^\pi dt = \frac{R\pi}{R^2 - 1}$$

$|x+y| \geq ||x| - |y||$

$\Rightarrow \int_{-\infty}^\infty \frac{\cos x}{1+x^2} = \frac{\pi}{e}$

$\downarrow R \rightarrow \infty$

~~ali~~

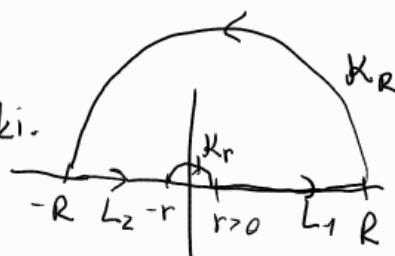
logariten

$$\ln z = \ln|z| + i \text{Arg}(z)$$

$\hookrightarrow$ kot

$$\ln: \underbrace{\mathbb{C} - (-\infty, 0]}_{\text{lahkotudi } (-\infty i, 0]} \rightarrow \mathbb{C}$$

2) Izračunaj $\int_0^\infty \frac{\ln x}{1+x^2} dx$. Uporabi zanko na sliki.



$$r < 1, R > 1$$

$$\ln: \mathbb{C} - (-\infty i, 0] \rightarrow \mathbb{C}$$

$$z, \text{Arg} \in (-\frac{\pi}{2}, \frac{3\pi}{2})$$

$$\gamma = L_1 \cup K_R \cup L_2 \cup K_r$$

$$f(z) = \frac{\ln z}{1+z^2}$$

$$\oint_\gamma \frac{\ln z}{1+z^2} dz = 2\pi i \text{Res}\left(\frac{\ln z}{1+z^2}, i\right) = 2\pi i \lim_{z \rightarrow i} \frac{\ln z}{z+i} = 2\pi i \frac{\ln i}{2i} = \pi [\ln 1 + i \frac{\pi}{2}] = i \frac{\pi^2}{2}$$

$$i \frac{\pi^2}{2} = \int_{L_1} \frac{\ln z}{1+z^2} dz + \int_{L_2} \frac{\ln z}{1+z^2} dz + \int_{K_r} \frac{\ln z}{1+z^2} dz + \int_{K_R} \frac{\ln z}{1+z^2} dz = \int_r^R \frac{\ln t}{1+t^2} dt + \int_r^R \frac{\ln t + i\pi}{1+t^2} dt + \int_{K_r} + \int_{K_R} =$$

$$= 2 \int_{-r}^R \frac{\ln t}{1+t^2} dt + \underbrace{i\pi \arctan t \Big|_{-r}^R}_{R \rightarrow \infty, r \rightarrow 0} + \int_{K_r} + \int_{K_R} \rightarrow \int_0^\infty \frac{\ln x}{1+x^2} dx = 0$$

$i\pi(\frac{\pi}{2} - 0) = i\frac{\pi^2}{2}$

Möbingsove transf.

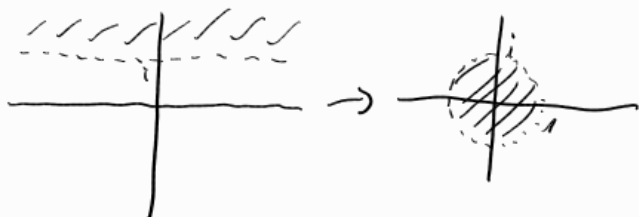
$$a, b, c, d \in \mathbb{C}$$

$$f(z) = \frac{az+b}{cz+d} : \mathbb{C} - \left\{ -\frac{d}{c} \right\} \rightarrow \mathbb{C}$$

Lastnosti:

- konformna (ohranja kote)
- slika premice in krožnice $\vee$ premice in krožnice
- določena s sliko treh točk
 $\rightarrow$ če želimo $f(z_1)=w_1, f(z_2)=w_2, f(z_3)=w_3 \Rightarrow \exists!$ Möb. tr.

1) poišči biholomorfno preslikavo $f: \{z \in \mathbb{C}; \operatorname{Im}(z) > 1\} \rightarrow B(0,1)$
odprt
krog



a) najprej $\{z \in \mathbb{C}; \operatorname{Im}(z) > 0\} \rightarrow B(0,1)$

Poizkusimo z Möb. tr.

$$f(0) = -1$$

$$f(z) = \frac{az+b}{cz+d}$$

$$f(0) = \frac{b}{d} = -1 \rightarrow b = -d$$

$$f(1) = -i$$

$$\frac{a}{c} = 1 \rightarrow a = c$$

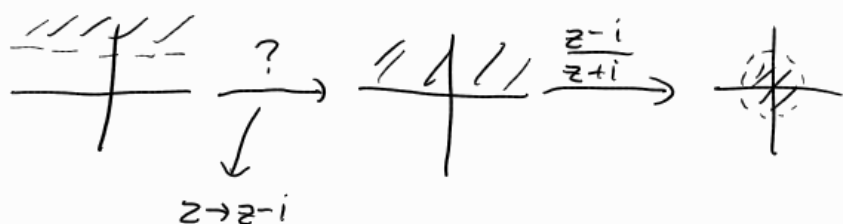
$$\rightarrow f(z) = \frac{az-d}{az+d} \quad f(1) = \frac{a-d}{a+d} = -i$$

$$a-d = -ia - id$$

$$a(1+i) = d(1-i)$$

$$d = a \frac{1+i}{1-i} = ai$$

$$f(z) = \frac{az-ia}{az+ia} = \frac{z-i}{z+i}$$



$$\rightarrow g(z) = \frac{(z-i)-i}{(z-i)+i} = \frac{z-2i}{z}$$